INCH-POUND
MIL-R-46837B (MI)
16 July 1990
SUPERSEDING
MIL-R-46837A
5 November 1986

MILITARY SPECIFICATION

RACEWAYS, STEEL, RIGID

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for rigid steel raceway conduits which consist of a base and cover so formed as to distinguish raceways from metallic tubular conduits.

1.2 Types and sizes. The raceways shall be of the following types and sizes, as specified (see 6.2):

Type I - Base snaps on cover as one unit - size no. 2,5,7, and 10 (see table I and figure 1).

Type II - Cover snaps over base as separate units - size no. 15 (see table II and figure 2).

Type III - Cover snaps over outside of base as separate units - size no. 19c and 19b (see table III and figure 3).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, Atten: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type IV - Cover snaps inside of base as separate units - size no. 21c, 21b, 30c, and 30b (see table IV and figure 4).

1.2.1 Color. The colors shall be identified in accordance with the following:

0 - Black	3 - White
1 - Brown	4 - Red

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Boxes, Shipping, Fiberboard
QQ-S-781	Strapping, Steel and Seals

MILITARY

MIL-P-116	Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

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MIL-STD-1190 Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

UNDERWRITER LABORATORIES, INC. STANDARD:

Subject no. 5 for Surface Metal - Raceways and Fittings.

(Application for copies should be addressed to Underwriter Laboratories, Inc., Walt Whitman Road, Melville, Long Island, New York 11749).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Raceways. Raceways covered by this specification shall conform to the requirements of Underwriter Laboratories, Inc. standard subject no. 5 for surface raceways and fittings.

3.2 Construction. Raceways shall be constructed of mild steel and shall consist of two metal pieces so formed and constructed as to provide a complete and secure enclosure which will afford adequate protection against mechanical injury or rodent destruction of wire conductors.

3.3 Dimensions. Overall dimensions for each type raceway shall be as specified in tables I, II, III, and IV, and figures 1, 2, 3, and 4. Radii and dimensions for controlling the general shape shall be controlled by the manufacturer's practice.

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3.4 Tolerances. Height and width tolerance shall be plus or minus 1/16 inch on references "X" and "Y" dimensions of tables I, II, III, and IV. The materials thickness tolerances shall be plus 0.010 and minus 0.005 inch, and the length shall be plus or minus 1/4 inch.

3.5 Marking. Each length of raceway shall be identified by the name or trademark of the manufacturer, this specification, type and size, and in accordance with MIL-STD-130.

TABLE I. Type I

Size	Dimensions			Length-Ft		Wire-Size	Race Capacity	
	X	Y	Z	Cover	Base	AWG	Type R or RH	Type T or RU
No 2	1/2	11/32	0.025	5	5	14 12	3 2	3 3
No. 5	3/4	17/32	0.040	10	10	14 12 10	4 3 2	6 6 4
No. 7	3/4	21/32	0.040	10	10	14 12 10 8	5 4 3 2	8 8 6 3
No. 10	1-5/16	15/16	0.040	10	10	14 12 10 8 6	10 10 6 5 4	10 10 8 8 5

Note: Wires are pulled through the raceways

Figure 1

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TABLE II. Type II

Size	Dimensions			Length-Ft		Wire Size	Race Capacity	
	X	Y	Z	Cover	Base	AWG	Type R or RH	Type T or RU
No. 15	1-9/16	11/32	0.040	10	10	14 12 10 8	4 4	8 6 4 4

Figure 2

TABLE III. Type III

Size	Dimensions			Length-Ft		Wire Size	Race Capacity	
	X	Y	Z	Cover	Base	AWG	Type R or RH	Type T or RU
No 19C Cover	13/16	--	0.025	5				
No.19 B (Base)	13/16	9/16	0.025	--	10	14 12	3 3	3 3

Note: Screw Knockouts are provided in base

Figure 3

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TABLE IV. Type IV

Size	Dimensions			Length-Ft		Wire Size	Race Capacity	
	X	Y	Z	Cover	Base	AWG	Type R or RH	Type T or RU
No.21 C Cover	1-1/14	---	0.040	5		14 12	10 4	10 6
No.21 B (Base	1-1/4	7/8	0.040	--	10	12	10	10
No.30 C Cover	2-3/4	---	0.040	10	--			
No.30 B	2-3/4	1-7/16	0.040	--	10	14 12 10	10 10 10	10 10 10

Note: Screw Knockouts are provided in base

Figure 4

3.6 Cleaning. The cleaning process shall leave the raceways in a condition such that the protective coating will adhere firmly to all surfaces.

3.7. Finish. Raceways shall be finished with baked enamel on all surfaces to provide protection from corrosion. the composition and application of enamel shall be in accordance with commercial practice.

3.7.1 Enamel coating. The enamel coating shall evenly cover the surface to which it is applied. The enamel coating shall withstand the test requirements specified herein (see 4.2.2.2).

3.7.2 Enamel color. Color of the baked enamel shall be as specified by the ordering data (see 6.2).

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3.8 Workmanship. Raceways surfaces shall be free of foreign matter, contain no sharp edges or burrs, and shall be smooth with uniform color. There shall be no cracks or splits in the material, however, slight enamel crazing or scoring on the raceway may be acceptable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. the Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary; to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspections system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to accept defective material.

4.2 Inspection.

4.2.1 Inspection lot. A lot, for inspection purposes, shall consist of raceways that are the same material, type, finish, size and manufactured under essentially the same conditions.

4.2.2 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be conducted in accordance with Table V. Quality conformance inspection shall be applied to each lot of completed items being offered for fulfillment of the contract. Quality conformance inspections shall consist of the following:

a. Examination (4.2.2.1)

b. Tests (4.2.2.2)

c. Packaging inspection (4.2.2.3)

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4.2.2.1 Examination. The raceways shall be examined for design, construction, materials, components, dimensions, marking, and workmanship to determine compliance with the requirements of this specification. Any deviation from those requirements shall be cause for rejection.

4.2.2.2 Tests. An enamel finished sample from each lot shall be tested with a 3/4 inch diameter ball at a 5 inch-pound impact over a hollow anvil. After the impact, the enamel coating shall remain unbroken, even if the surface of the metal is dented by the impact force.

4.2.2.3 Packaging inspection. Preservation, packaging, packing, and marking for shipment and storage shall be inspected to determine compliance with the requirements of section 5.

TABLE V. Sampling plan.

Lot Size	Sample Size
2 to 3	100%
4 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1200	15
1201 to 3200	18
3201 to 10,000	22
10,001 and over	29
In all cases: Acceptance number is ZERO Rejection number is ONE	

5. PACKAGING

5.1 Preservation. Preservation, packaging, packing, unitization, and marking shall be level A, B, or Commercial (see 6.2).

5.1.1 Level A or B. Raceways shall be packaged in accordance with MIL-P-116, Method III, utilizing containers conforming to PPP-B-636, class weather resistant or PPP-B-636, variety 2. Raceways shall be packaged in 5 or 10 increments, a maximum of 100 feet to the container.

5.1.2 Commercial. Preservation and packaging shall be in accordance with the requirements of MIL-STD-1190.

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5.2 Packing.

5.2.1 Level A. Packaged raceways shall be packed in containers conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specification, except strapping shall conform to QQ-S-781 type I or IV, finish A.

5.2.2 Level B. Packaged raceways shall be packed in fiberboard, containers conforming to PPP-B-636 class weather resistant, style optional, closures shall be in accordance with the appendix thereto.

5.2.3 Commercial. Packaged raceways shall be packed in accordance with the requirements of MIL-STD-1190.

5.3 Marking.

5.3.1. Level A or B. Marking shall be in accordance with MIL-STD-129. Additional markings shall include the following:

a. Specification number, type and size

b. Quantity contained (as defined in contract).

c. Name of manufacturer.

d. contract or purchase order number.

5.3.2. Commercial. Marking shall be in accordance with MIL-STD-1190. Additional markings shall include the following:

a. Specification number, type and size

\ b. Quantity contained (as defined in contract).

c. Name of manufacturer.

d. contract or purchase order number.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. Conduit raceways procured under this specification are intended to contain and protect electrical wires, cables or harness on floor, walls, and ceiling surfaces from mechanical or rodent damages.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. Type and size of raceways (see 1.2).

d. Number of feet in carton (see 5.1.2)

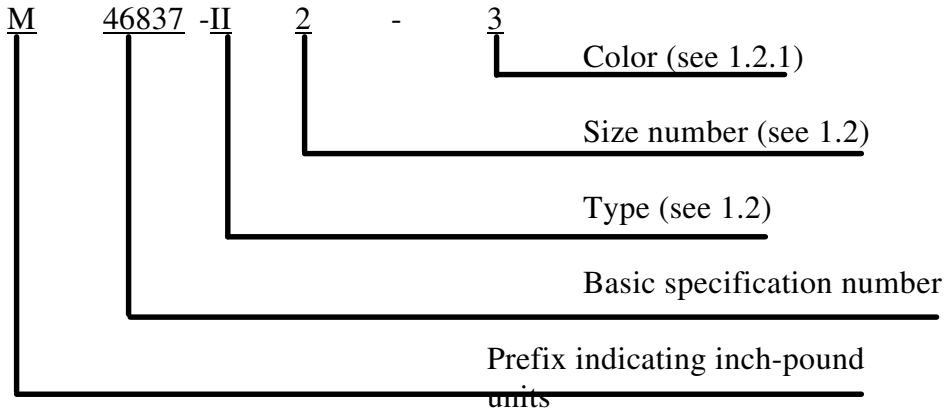
e. Length if different than specified in 5.1.1.

f. Color required (see 1.2.1 and 3.7.2).

g. If Government is to perform tests (see 4.1).

h. Variance in testing (see 4.2.2).

6.3 Part of Identifying Number (PIN). The PIN should consist of the letter M, the basis number of this specification, and dash number denoting type and size number.



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6.4 Subject term (key word) listing.

Conduits
Enamel
Snaps, cover

6.5 Changes from previous issue**.** Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing Activity:
Army - MI

Review Activities:
DLA - GS

Project No. 5975-A002

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